Secure Deployment Documentation – Nilavan Application

Project Overview

This document explains the complete deployment process of a secure production-ready Next.js application on Ubuntu 24.04 using:

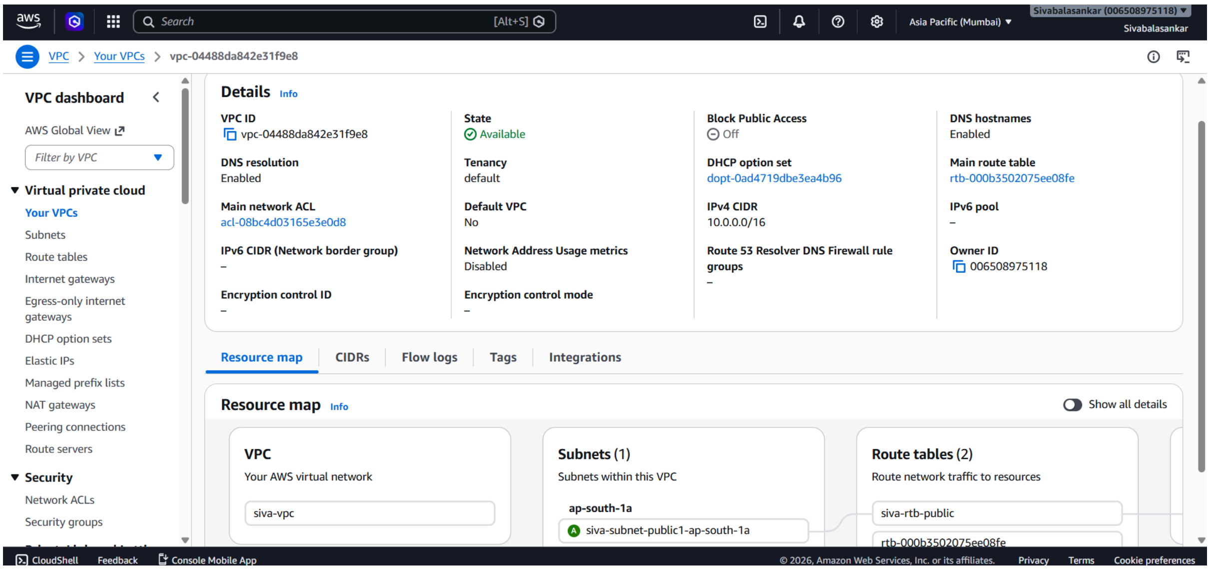
- Ubuntu 24.04 VPS
- Nginx Reverse Proxy
- PM2 Process Manager
- DuckDNS
- SSL using Certbot
- SSH Hardening
- GitHub SSH Deployment
- Elastic IP Configuration

1. Infrastructure Setup

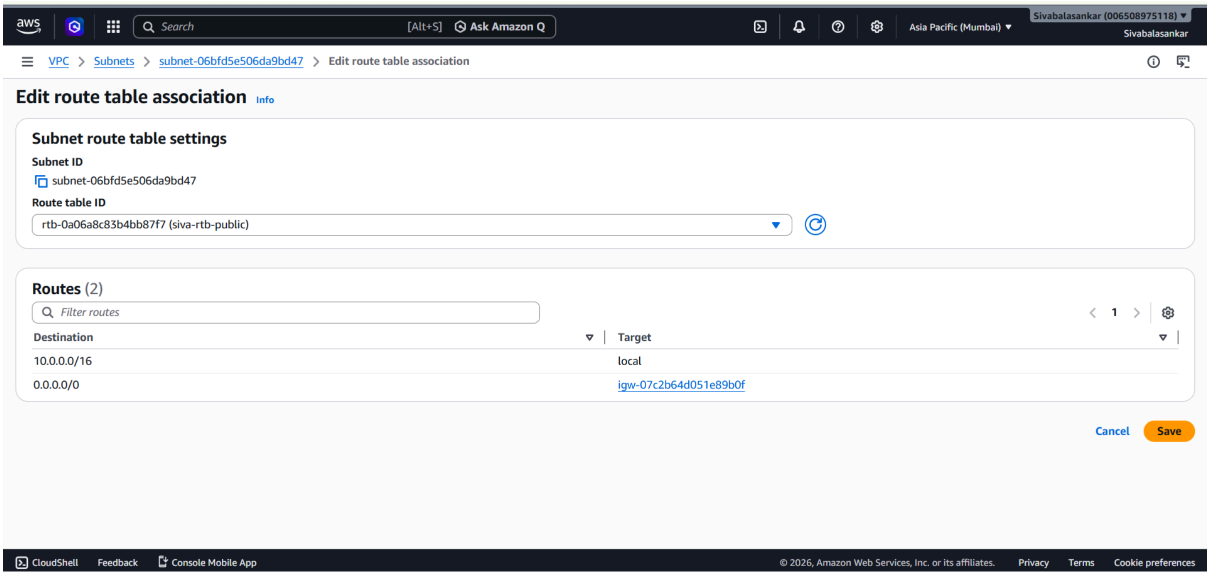
1.1 Created VPC Network

A custom VPC was created with:

- Public subnet
- Internet Gateway (IGW)
- Route table configuration
- Public internet access



- VPC Dashboard
- Subnet Configuration
- Route Table
- Internet Gateway Attached

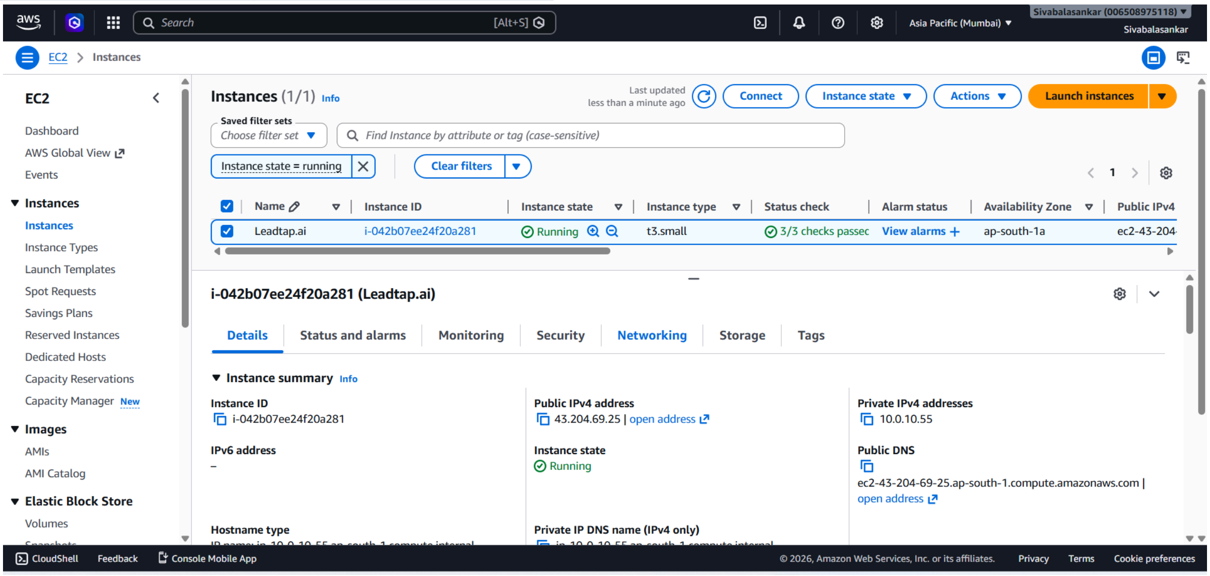


2. Virtual Machine Creation

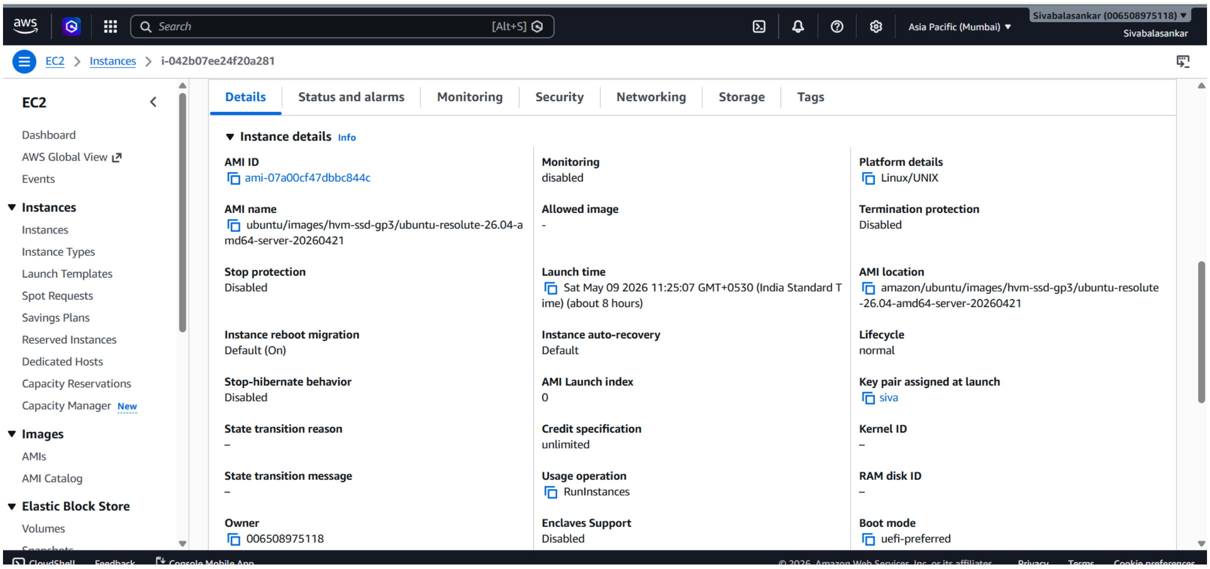
2.1 Created Ubuntu VM

VM Details:

Configuration	Value
OS	Ubuntu 24.04 LTS
Instance Type	e2-medium
Public IP	Attached
SSH Access	Enabled



- VM instance dashboard
- Running status
- External IP
- Machine type



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3. Elastic IP Configuration

3.1 Created and Attached Elastic IP

An Elastic IP was created from the AWS Management Console and attached to the EC2 instance to provide a static public IP address.

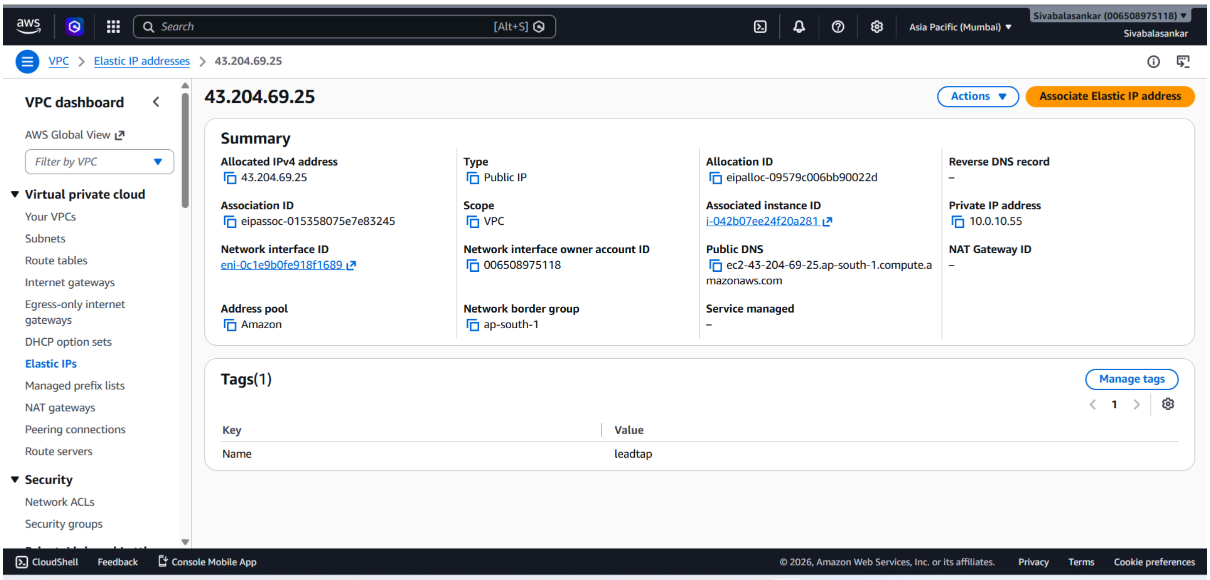
Steps Performed in AWS Portal

1. Logged in to the AWS Management Console.
2. Navigated to:

`EC2 Dashboard → Network & Security → Elastic IPs`

3. Clicked on: `Allocate Elastic IP addres`
4. Clicked: `Allocate`
5. Selected the newly created Elastic IP.
6. Clicked: `Actions → Associate Elastic IP address`
7. Chose:
   - Resource type: Instance
   - Selected the Ubuntu EC2 instance
   - Selected the private IP
8. Clicked: `Associate`

The Elastic IP was later mapped to the DuckDNS domain.



- Elastic IP allocation page
- Elastic IP attached to EC2 instance
- AWS EC2 dashboard showing Elastic IP association
- Later elastic IP is mapped with DuckDNS

4. User Creation & Hardening

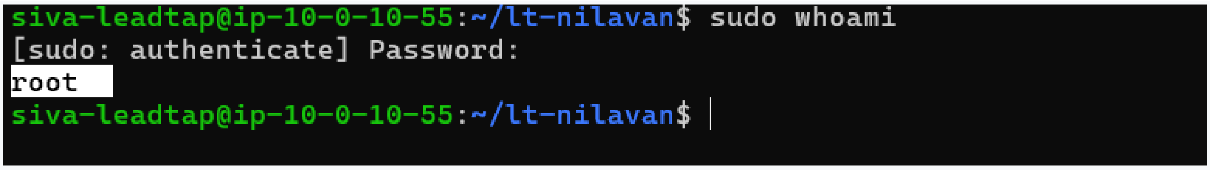
4.1 Created Non-Root User

A dedicated deployment user was created.

Commands Used

```
sudo adduser siva-leadtap
sudo usermod -aG sudo siva-leadtap
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
tss:x:987:987:tss user for tpm2:/:/usr/sbin/nologin
uuidd:x:101:103::/run/uuidd:/usr/sbin/nologin
sshd:x:986:65534:sshd user:/run/sshd:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
tcpdump:x:985:985:tcpdump:/nonexistent:/usr/sbin/nologin
landscape:x:103:106::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:984:984:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:983:983:User for polkitd:/:/usr/sbin/nologin
ec2-instance-connect:x:104:65534::/nonexistent:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
siva-leadtap:x:1001:1001:siva-leadtap,,,:/home/siva-leadtap:/bin/bash
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$
```



```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ sudo whoami
[sudo: authenticate] Password:
root
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$
```

---

5. SSH Security Hardening

5.1 SSH Configuration

SSH was hardened using the following settings:

Setting	Value
SSH Port	2222
Root Login	Disabled
Password Authentication	Disabled
Public Key Authentication	Enabled

SSH Config File

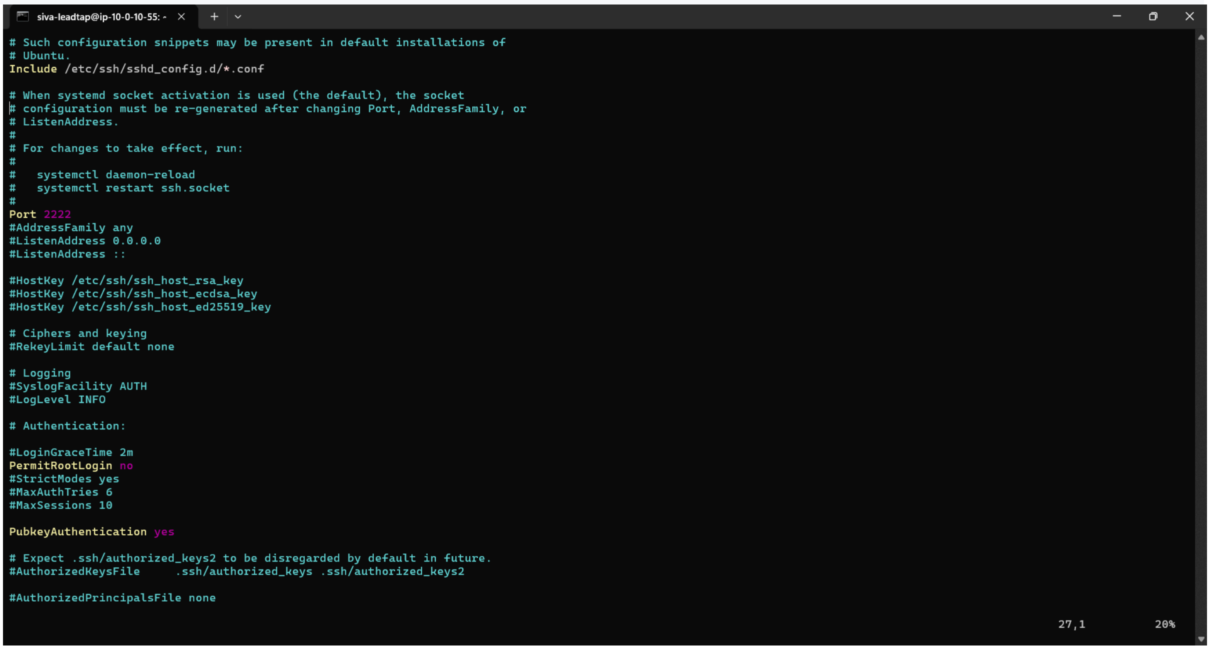
```
sudo nano /etc/ssh/sshd_config
```

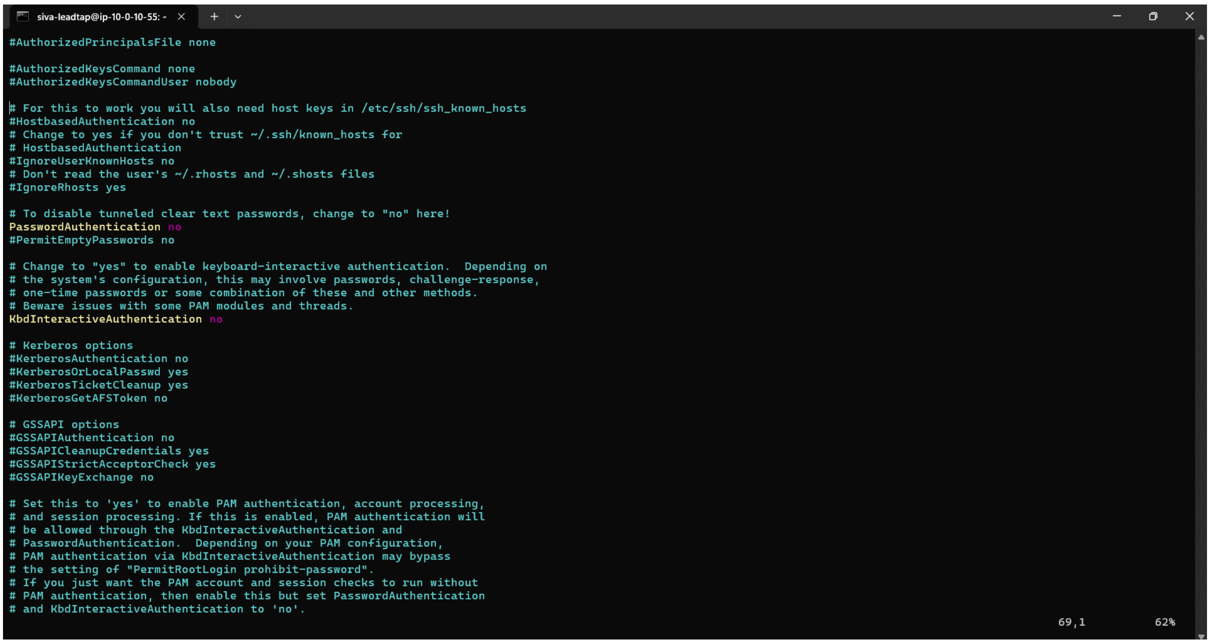
Configuration Added

```
Port 2222
PermitRootLogin no
PasswordAuthentication no
PubkeyAuthentication yes
```

Restart SSH

```
sudo systemctl restart ssh
```





sudo systemctl status ssh

```
ubuntu@ip-10-0-10-55:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
    Drop-In: /usr/lib/systemd/system/ssh.service.d
             └─ec2-instance-connect.conf
     Active: active (running) since Sat 2026-05-09 05:55:21 UTC; 22h ago
 Invocation: 225611cbb2934d6fbbd4e62d3c036779
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 992 (sshd)
      Tasks: 1 (limit: 1741)
     Memory: 7.8M (peak: 30.5M)
        CPU: 2.334s
     CGroup: /system.slice/ssh.service
             └─992 "sshd: /usr/sbin/sshd -D -o AuthorizedKeysCommand /usr/share/ec2-instance-connect/eic_run_authorized>

May 09 13:34:47 ip-10-0-10-55 sshd-session[16418]: Accepted publickey for siva-leadtap from 157.50.1.159 port 47118 ssh>
May 09 13:34:47 ip-10-0-10-55 sshd-session[16418]: pam_unix(sshd:session): session opened for user siva-leadtap(uid=100>
May 09 14:13:01 ip-10-0-10-55 sshd-session[17246]: Accepted publickey for siva-leadtap from 157.50.7.124 port 36981 ssh>
May 09 14:13:01 ip-10-0-10-55 sshd-session[17246]: pam_unix(sshd:session): session opened for user siva-leadtap(uid=100>
May 09 20:14:43 ip-10-0-10-55 sshd-session[19965]: Invalid user  from 18.97.19.141 port 34520
May 09 20:14:44 ip-10-0-10-55 sshd-session[19965]: Connection closed by invalid user  18.97.19.141 port 34520 [preauth]
May 10 04:12:24 ip-10-0-10-55 sshd-session[21073]: Accepted publickey for ubuntu from 157.50.1.143 port 46157 ssh2: RSA>
May 10 04:12:24 ip-10-0-10-55 sshd-session[21073]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by >
May 10 04:30:27 ip-10-0-10-55 sshd-session[21378]: Accepted publickey for ubuntu from 157.50.1.143 port 45160 ssh2: RSA>
May 10 04:30:27 ip-10-0-10-55 sshd-session[21378]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by >
lines 1-26/26 (END)
```

ss -tulpn | grep 2222

```
siva-leadtap@ip-10-0-10-55:~$ netstat -tupln | grep 2222
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:2222            0.0.0.0:*               LISTEN      -
tcp6       0      0 :::2222                 :::*                    LISTEN      -
siva-leadtap@ip-10-0-10-55:~$
```

sudo cat /etc/ssh/sshd_config

---

6. SSH Key Authentication

6.1 Added Public Key Authentication

The SSH public key was added to the new deployment user.

Commands Used

```
mkdir -p ~/.ssh
nano ~/.ssh/authorized_keys
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
```

```
siva-leadtap@ip-10-0-10-55:~$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDp3rjggZcgpSI2asMIdcobXJ81OISsslXE74m5wzx+5JVrzWblvgFzLJyUCZwZcVm+lO1/xOGFqK6CFlHjjTHdSnwXe0Xd6fU1KXi4czF05t4gSOQumrYB
fXQrFnMR4N028PCwrOMuXzlKA8zvjX3EnRmKlrTiAR4RJwX3qwY1Ch2BIDdOURIitYiKHy6igNoAngaM2/Qgha2XIIf7wDRObnJ7q+AFMGfMAVQRtIiTX6ZOMQiUcI8SfolyvK30tHBXztQ5YKQb1SQRl9iH
IlvQst[redacted] siva
siva-leadtap@ip-10-0-10-55:~$
```

```
siva-leadtap@ip-10-0-10-55:~$ ls -la ~/.ssh/authorized_keys
-rw------- 1 siva-leadtap siva-leadtap 386 May  8 10:26 /home/siva-leadtap/.ssh/authorized_keys
siva-leadtap@ip-10-0-10-55:~$
```

7. Installed Required Software

7.1 Installed Git

```
sudo apt update
sudo apt install git -y
```

7.2 Installed Node.js 20

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install nodejs -y
```

7.3 Installed PM2

```
sudo npm install -g pm2
```

8. Deployment

8.1 Cloned Repository using SSH

Commands Used

```
git clone git@github.com:Leadtap/lt-nilavan.git
```

```
siva-leadtap@ip-10-0-10-55:~$ ssh -T git@github.com
Hi Sivabalasankar2002! You've successfully authenticated, but GitHub does not provide shell access.
siva-leadtap@ip-10-0-10-55:~$ |
```

```
siva-leadtap@ip-10-0-10-55:~$ cd lt-nilavan/
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ ls
README.md  components       hooks  next-env.d.ts      package.json        public  tailwind.config.ts
app        components.json  lib    package-lock.json  postcss.config.mjs  styles  tsconfig.json
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ |
```

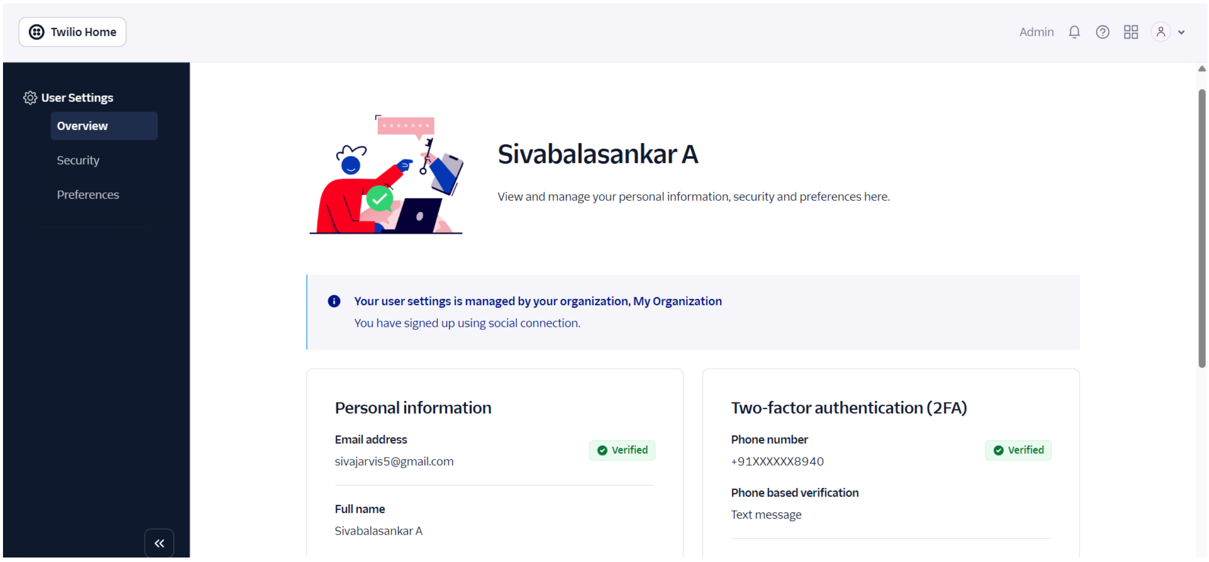
SendGrid Email Service Configuration

Created SendGrid Account and API Key

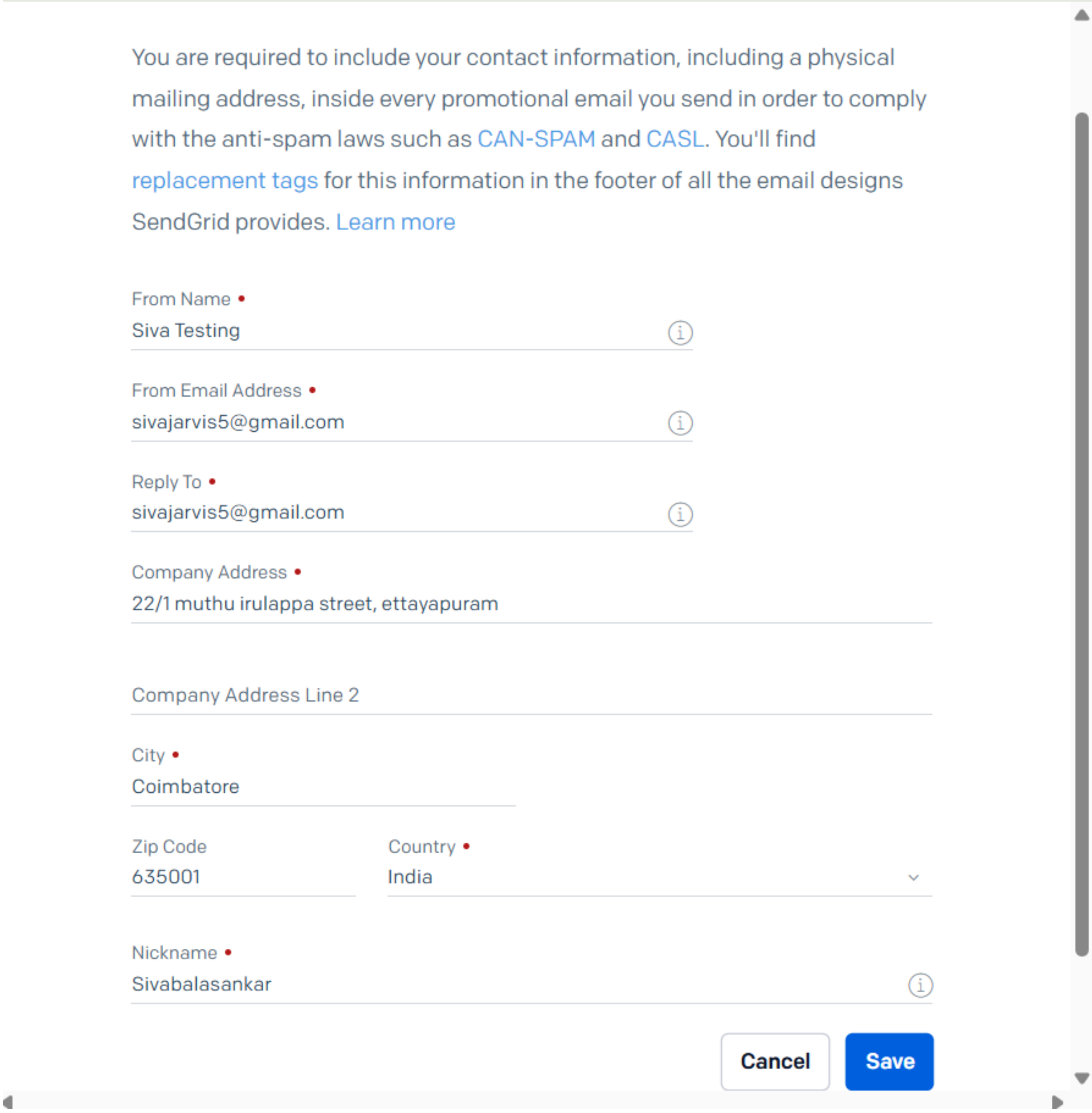
A SendGrid account was created for handling email functionality from the Next.js application.

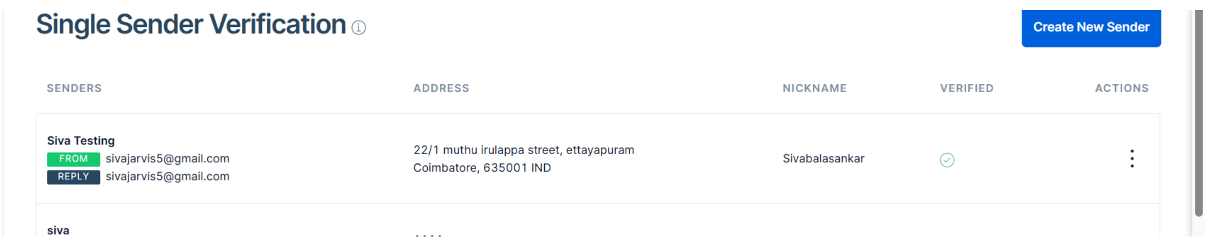
The following steps were completed:

Created SendGrid account

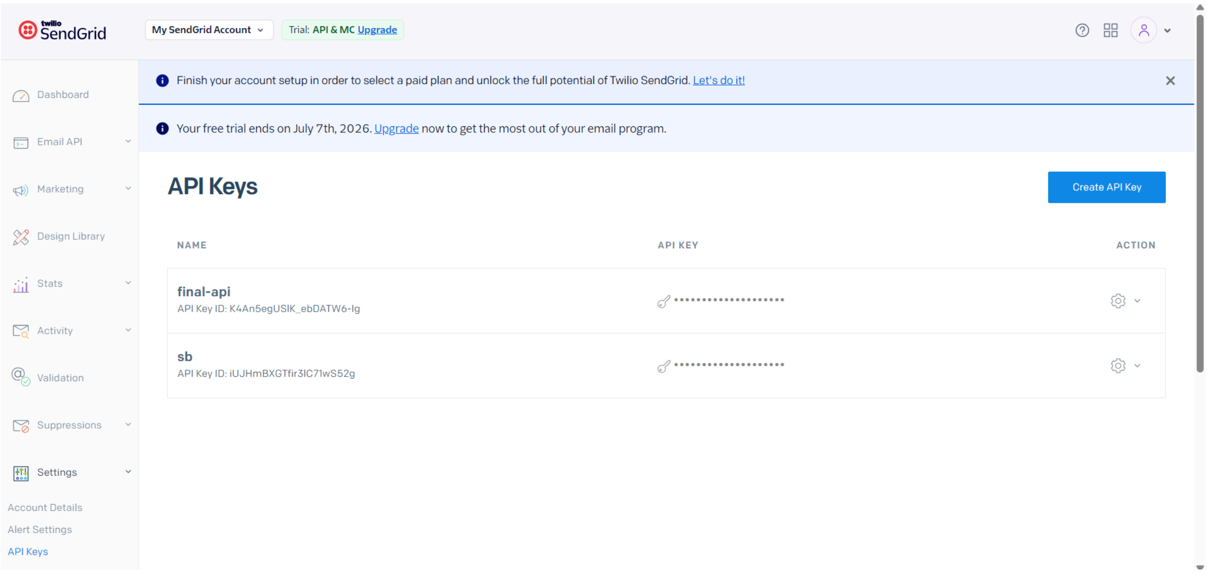


Verified sender email address





Generated SendGrid API Key



Added environment variables in .env.local

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ ls -la
total 280
drwxrwxr-x  14 siva-leadtap siva-leadtap   4096 May  9 14:27 .
drwxr-x---  10 siva-leadtap siva-leadtap   4096 May  9 14:27 ..
-rw-rw-r--   1 siva-leadtap siva-leadtap    127 May  8 17:10 .env.local
-rw-r--r--   1 siva-leadtap siva-leadtap  12288 May  9 11:49 .env.local.swp
drwxrwxr-x   7 siva-leadtap siva-leadtap   4096 May  9 13:01 .git
drwxr-xr-x   3 siva-leadtap siva-leadtap   4096 May  9 13:51 .next
drwxrwxr-x   4 siva-leadtap siva-leadtap   4096 May  8 10:31 .npm
drwxrwxr-x   8 siva-leadtap siva-leadtap   4096 May  8 10:31 .nvm
drwxrwxr-x   5 siva-leadtap siva-leadtap   4096 May  9 05:55 .pm2
-rw-rw-r--   1 siva-leadtap siva-leadtap     18 May  8 10:39 README.md
drwxrwxr-x   3 siva-leadtap siva-leadtap   4096 May  8 10:39 app
drwxrwxr-x   3 siva-leadtap siva-leadtap   4096 May  8 10:39 components
-rw-rw-r--   1 siva-leadtap siva-leadtap    443 May  8 10:39 components.json
drwxrwxr-x   2 siva-leadtap siva-leadtap   4096 May  8 10:39 hooks
drwxrwxr-x   2 siva-leadtap siva-leadtap   4096 May  8 10:39 lib
-rw-rw-r--   1 siva-leadtap siva-leadtap    262 May  9 11:39 next-env.d.ts
drwxrwxr-x 211 siva-leadtap siva-leadtap  12288 May  9 11:38 node_modules
-rw-rw-r--   1 siva-leadtap siva-leadtap 173398 May  9 11:38 package-lock.json
-rw-rw-r--   1 siva-leadtap siva-leadtap   2305 May  9 11:38 package.json
-rw-rw-r--   1 siva-leadtap siva-leadtap    135 May  8 10:39 postcss.config.mjs
drwxrwxr-x   3 siva-leadtap siva-leadtap   4096 May  8 10:39 public
drwxrwxr-x   2 siva-leadtap siva-leadtap   4096 May  8 10:39 styles
-rw-rw-r--   1 siva-leadtap siva-leadtap   2961 May  8 10:39 tailwind.config.ts
-rw-rw-r--   1 siva-leadtap siva-leadtap    595 May  8 10:39 tsconfig.json
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$
```

9. Application Build & PM2 Setup

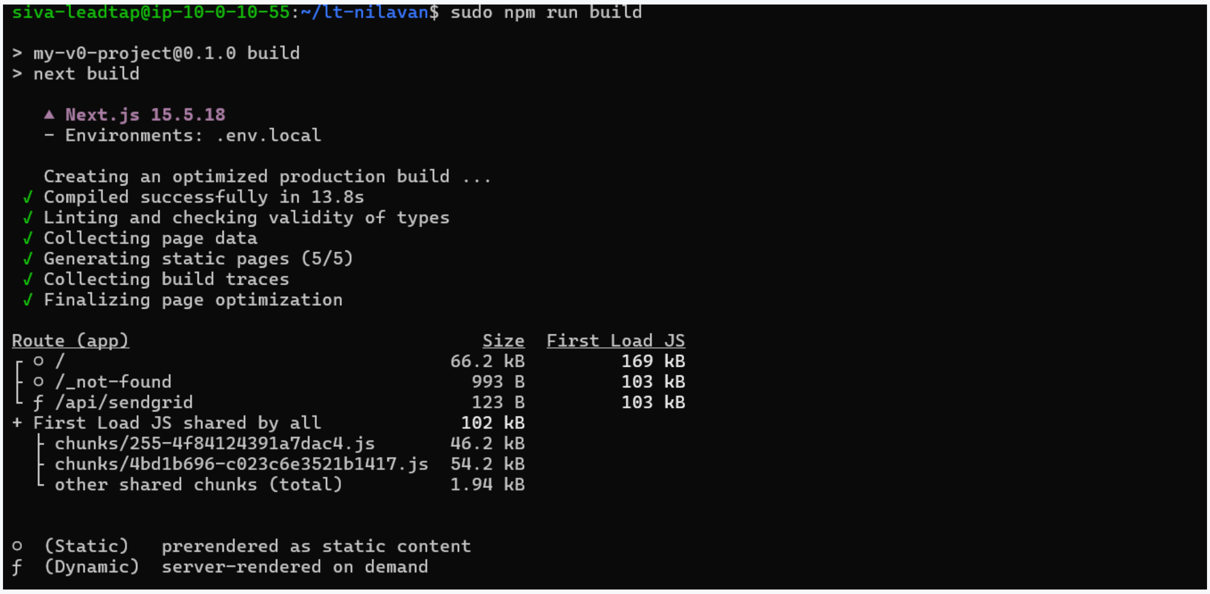
9.1 Installed Dependencies

```
npm install
```

```
README.md  components       hooks  next-env.d.ts  package-lock.json  postcss.config.mjs  styles              tsconfig.json
app        components.json  lib    node_modules   package.json       public              tailwind.config.ts
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ npm install

up to date, audited 301 packages in 3s

52 packages are looking for funding
  run `npm fund` for details

2 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.14.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.14.1
npm notice To update run: npm install -g npm@11.14.1
npm notice
```

9.2 Built Next.js Application

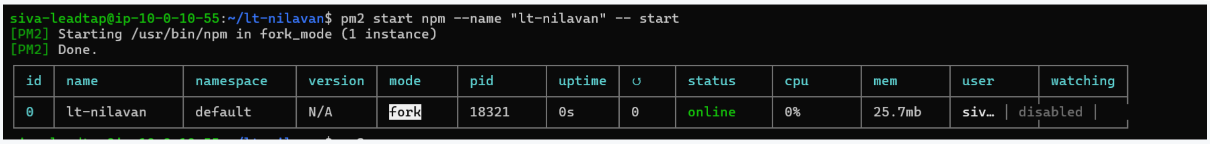
```
npm run build
```



```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ sudo npm run build

> my-v0-project@0.1.0 build
> next build

   ▲ Next.js 15.5.18
   - Environments: .env.local

   Creating an optimized production build ...
 ✓ Compiled successfully in 13.8s
 ✓ Linting and checking validity of types
 ✓ Collecting page data
 ✓ Generating static pages (5/5)
 ✓ Collecting build traces
 ✓ Finalizing page optimization

Route (app)                              Size  First Load JS
┌ ○ /                                  66.2 kB         169 kB
├ ○ /_not-found                          993 B         103 kB
└ ƒ /api/sendgrid                        123 B         103 kB
+ First Load JS shared by all           102 kB
  ├ chunks/255-4f84124391a7dac4.js     46.2 kB
  ├ chunks/4bd1b696-c023c6e3521b1417.js 54.2 kB
  └ other shared chunks (total)        1.94 kB


○  (Static)   prerendered as static content
ƒ  (Dynamic)  server-rendered on demand
```

9.3 Started Application with PM2

```
pm2 start npm --name "lt-nilavan" -- start
```

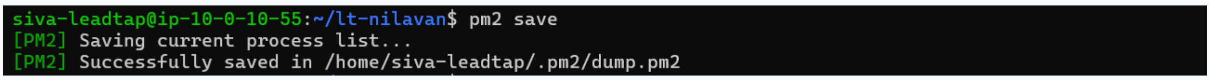


```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ pm2 start npm --name "lt-nilavan" -- start
[PM2] Starting /usr/bin/npm in fork_mode (1 instance)
[PM2] Done.
```

id	name	namespace	version	mode	pid	uptime	↺	status	cpu	mem	user	watching
0	lt-nilavan	default	N/A	fork	18321	0s	0	online	0%	25.7mb	siv…	disabled

9.4 Saved PM2 Processes

```
pm2 save
```



```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ pm2 save
[PM2] Saving current process list...
[PM2] Successfully saved in /home/siva-leadtap/.pm2/dump.pm2
```

9.5 Enabled PM2 Auto Startup

```
pm2 startup
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ pm2 startup
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u siva-leadtap --hp /home/siva-leadtap
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u siva-leadtap --hp /home/siva-lead
tap
[PM2] Init System found: systemd
Platform systemd
Template
[Unit]
Description=PM2 process manager
Documentation=https://pm2.keymetrics.io/
After=network.target

[Service]
Type=forking
User=siva-leadtap
LimitNOFILE=infinity
LimitNPROC=infinity
LimitCORE=infinity
Environment=PATH=/home/siva-leadtap/.nvm/versions/node/v20.20.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games
:/snap/bin:/usr/bin:/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
Environment=PM2_HOME=/home/siva-leadtap/.pm2
PIDFile=/home/siva-leadtap/.pm2/pm2.pid
Restart=on-failure

ExecStart=/usr/lib/node_modules/pm2/bin/pm2 resurrect
ExecReload=/usr/lib/node_modules/pm2/bin/pm2 reload all
ExecStop=/usr/lib/node_modules/pm2/bin/pm2 kill
```

10. Nginx Reverse Proxy Configuration

10.1 Installed Nginx

```
sudo apt install nginx -y
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ sudo systemctl status nginx.service
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-05-09 11:46:24 UTC; 2h 49min ago
 Invocation: 64ad8c9881784247ab21de82062243ea
       Docs: man:nginx(8)
   Main PID: 11831 (nginx)
      Tasks: 3 (limit: 1741)
     Memory: 8.6M (peak: 30.4M)
        CPU: 229ms
     CGroup: /system.slice/nginx.service
             ├─11831 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─11832 "nginx: worker process"
             └─11833 "nginx: worker process"

May 09 11:46:24 ip-10-0-10-55 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
May 09 11:46:24 ip-10-0-10-55 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$
```

```
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$ nginx -v
nginx version: nginx/1.28.3 (Ubuntu)
siva-leadtap@ip-10-0-10-55:~/lt-nilavan$
```

10.2 Created Reverse Proxy Configuration

Configuration File

```
sudo nano /etc/nginx/sites-available/lt-nilavan
```

Nginx Configuration

server { listen 80; listen [::]:80;

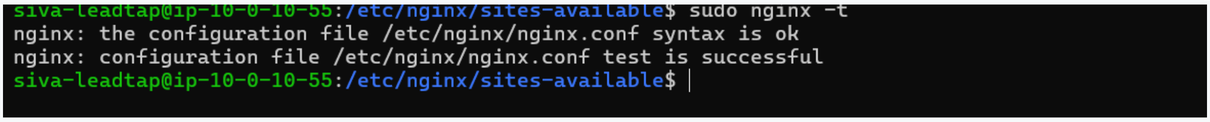
```
server_name _;

return 301 https://$host$request_uri;

}
```

Enable Site

```
sudo ln -s /etc/nginx/sites-available/lt-nilavan /etc/nginx/sites-enabled/
```

Test Nginx

```
sudo nginx -t
```



Restart Nginx

```
sudo systemctl restart nginx
```

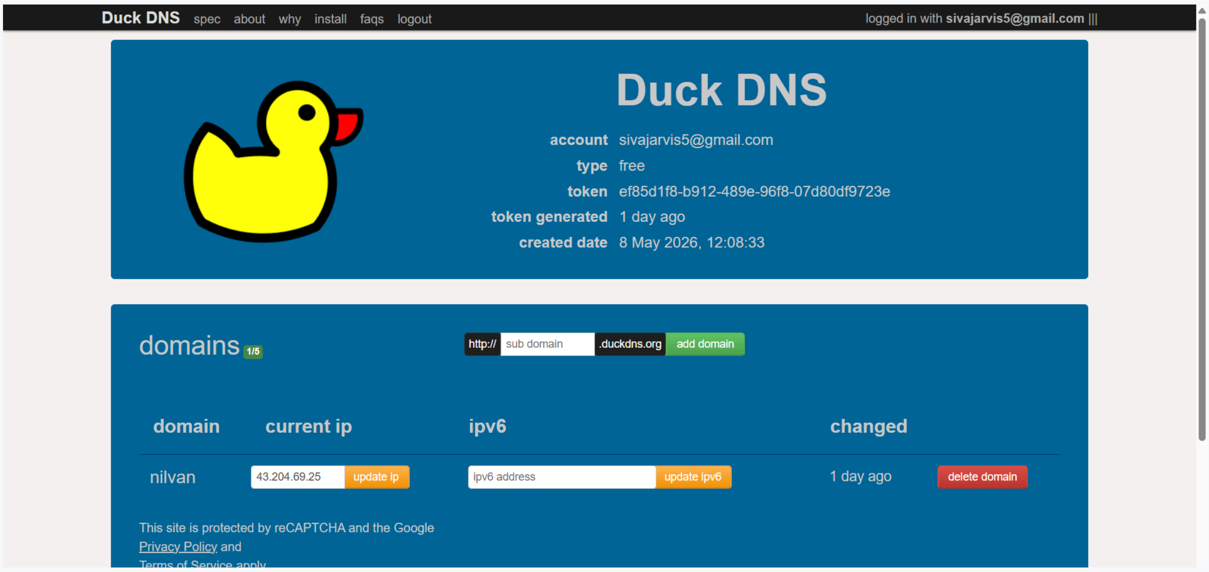
11. DuckDNS Configuration

11.1 Configured Domain

DuckDNS domain was mapped to the Elastic IP attached to the VPS.

Domain

```
https://nilvan.duckdns.org
```




```
nslookup nilvan.duckdns.org
```

```
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ nslookup nilvan.duckdns.org
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   nilvan.duckdns.org
Address: 43.204.69.25

siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$
```

---

12. SSL Configuration using Certbot

12.1 Installed Certbot

```
sudo apt install certbot python3-certbot-nginx -y
```

12.2 Generated SSL Certificate

```
sudo certbot --nginx -d nilvan.duckdns.org
```

```
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ curl -I https://nilvan.duckdns.org
HTTP/2 200
server: nginx
date: Sat, 09 May 2026 14:40:36 GMT
content-type: text/html; charset=utf-8
content-length: 47835
vary: rsc, next-router-state-tree, next-router-prefetch, next-router-segment-prefetch, Accept-Encoding
x-nextjs-cache: HIT
x-nextjs-prerender: 1
x-nextjs-prerender: 1
x-nextjs-stale-time: 300
x-powered-by: Next.js
cache-control: s-maxage=31536000
etag: "xdmb4g0t10wo"
x-frame-options: DENY
x-content-type-options: nosniff
referrer-policy: strict-origin-when-cross-origin
strict-transport-security: max-age=63072000; includeSubDomains; preload
permissions-policy: geolocation=(), microphone=(), camera=()
content-security-policy: default-src 'self'; base-uri 'self'; object-src 'none'; frame-ancestors 'none'; form-action 'self'; script-src 'self' 'unsafe-inlin
e' 'unsafe-eval' https:; style-src 'self' 'unsafe-inline' https:; img-src 'self' data: blob: https:; font-src 'self' data: https:; connect-src 'self' https:
; frame-src 'self' https:;
```

```
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ sudo certbot certificates
Saving debug log to /var/log/letsencrypt/letsencrypt.log

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Found the following certs:
  Certificate Name: nilvan.duckdns.org
    Serial Number: 56e5d714bac06e44df73b01bbcd27943279
    Key Type: ECDSA
    Domains: nilvan.duckdns.org
    Expiry Date: 2026-08-06 12:03:07+00:00 (VALID: 88 days)
    Certificate Path: /etc/letsencrypt/live/nilvan.duckdns.org/fullchain.pem
    Private Key Path: /etc/letsencrypt/live/nilvan.duckdns.org/privkey.pem
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
siva-leadtap@ip-10-0-10-55:/etc/nginx/sites-available$ |
```

13. HTTPS Nginx Configuration

13.1 Added SSL Configuration

HTTPS Nginx Configuration

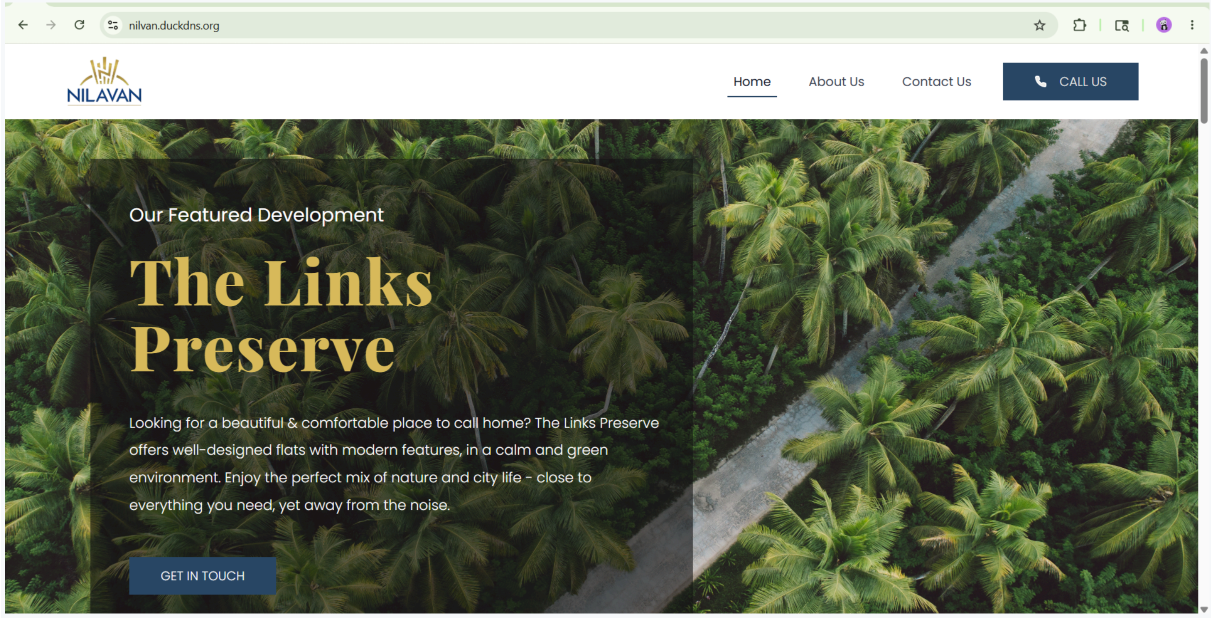
```
server {
    listen 443 ssl;
    listen [::]:443 ssl;

    http2 on;

    server_name nilvan.duckdns.org;

    # SSL
    ssl_certificate /etc/letsencrypt/live/nilvan.duckdns.org/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/nilvan.duckdns.org/privkey.pem;

    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers on;

    # Security headers
    add_header X-Frame-Options "DENY" always;
    add_header X-Content-Type-Options "nosniff" always;
    add_header Referrer-Policy "strict-origin-when-cross-origin" always;
    add_header Strict-Transport-Security "max-age=63072000;
includeSubDomains; preload" always;
```

```
    add_header Permissions-Policy "geolocation=(), microphone=(), camera=()"
always;
    add_header Content-Security-Policy "default-src 'self'; base-uri 'self';
object-src 'none'; frame-ancestors 'none'; form-action 'self'; script-src
'self' 'unsafe-inline' 'unsafe-eval' https:; style-src 'self' 'unsafe-inline'
https:; img-src 'self' data: blob: https:; font-src 'self' data: https:;
connect-src 'self' https:; frame-src 'self' https:;" always;

    # Block hidden files
    location ~ /\.(?!well-known).* {
        deny all;
        access_log off;
        log_not_found off;
    }

    # Block .git access
    location /.git {
        deny all;
        return 403;
    }

    # API rate limiting
    location /api/ {
        limit_req zone=api_limit burst=1 nodelay;

        proxy_pass http://127.0.0.1:3000;

        proxy_http_version 1.1;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto https;

        # Mitigate Next.js middleware vulnerability
        proxy_set_header x-middleware-subrequest "";

        proxy_cache_bypass $http_upgrade;
    }

    # Main app
    location / {
        proxy_pass http://127.0.0.1:3000;

        proxy_http_version 1.1;

        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto https;
```

```
        proxy_set_header x-middleware-subrequest "";

        proxy_cache_bypass $http_upgrade;
    }
}
```

Restart Nginx

```
sudo nginx -t
sudo systemctl restart nginx
```



---

14. Security Improvements

Implemented Security Controls

Security Feature	Status
Non-root deployment user	Enabled
SSH Port Changed	Enabled
Root Login Disabled	Enabled
Password Authentication Disabled	Enabled
Public Key Authentication	Enabled
HTTPS Enabled	Enabled
Reverse Proxy Configured	Enabled
PM2 Auto Start	Enabled
.git Public Access Blocked	Enabled
Elastic IP Configured	Enabled

15. Final Deployment Verification

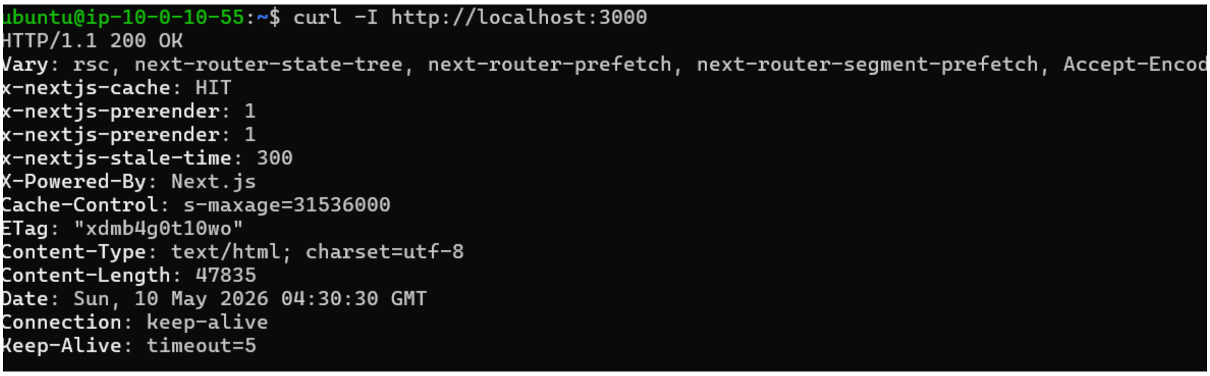
Production URL

`https://nilvan.duckdns.org`

Website Verification

Check Local App

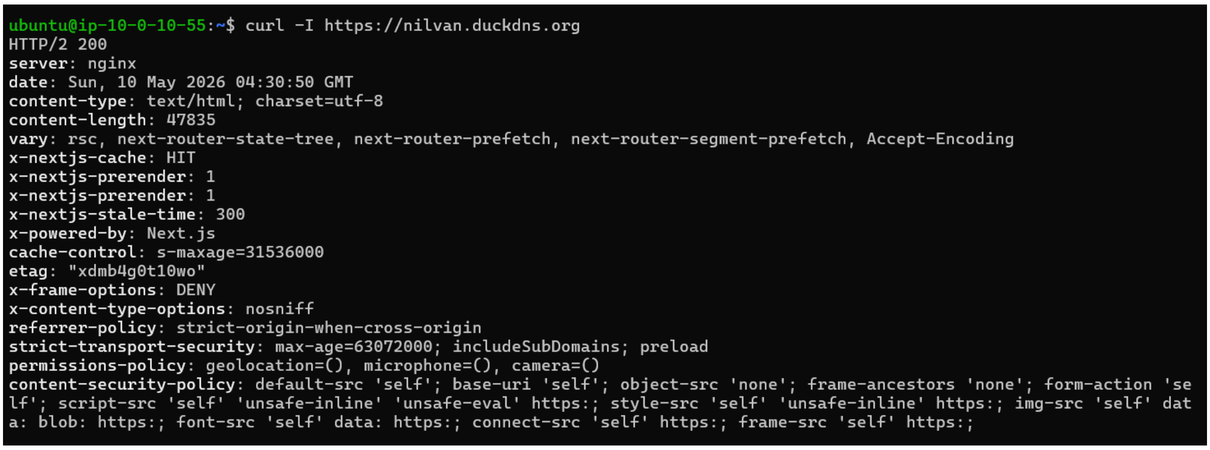
```
curl -I http://localhost:3000
```



```
ubuntu@ip-10-0-10-55:~$ curl -I http://localhost:3000
HTTP/1.1 200 OK
Vary: rsc, next-router-state-tree, next-router-prefetch, next-router-segment-prefetch, Accept-Encod
x-nextjs-cache: HIT
x-nextjs-prerender: 1
x-nextjs-prerender: 1
x-nextjs-stale-time: 300
X-Powered-By: Next.js
Cache-Control: s-maxage=31536000
ETag: "xdmb4g0t10wo"
Content-Type: text/html; charset=utf-8
Content-Length: 47835
Date: Sun, 10 May 2026 04:30:30 GMT
Connection: keep-alive
Keep-Alive: timeout=5
```

Check HTTPS Response

```
curl -I https://nilvan.duckdns.org
```



```
ubuntu@ip-10-0-10-55:~$ curl -I https://nilvan.duckdns.org
HTTP/2 200
server: nginx
date: Sun, 10 May 2026 04:30:50 GMT
content-type: text/html; charset=utf-8
content-length: 47835
vary: rsc, next-router-state-tree, next-router-prefetch, next-router-segment-prefetch, Accept-Encoding
x-nextjs-cache: HIT
x-nextjs-prerender: 1
x-nextjs-prerender: 1
x-nextjs-stale-time: 300
x-powered-by: Next.js
cache-control: s-maxage=31536000
etag: "xdmb4g0t10wo"
x-frame-options: DENY
x-content-type-options: nosniff
referrer-policy: strict-origin-when-cross-origin
strict-transport-security: max-age=63072000; includeSubDomains; preload
permissions-policy: geolocation=(), microphone=(), camera=()
content-security-policy: default-src 'self'; base-uri 'self'; object-src 'none'; frame-ancestors 'none'; form-action 'se
lf'; script-src 'self' 'unsafe-inline' 'unsafe-eval' https:; style-src 'self' 'unsafe-inline' https:; img-src 'self' dat
a: blob: https:; font-src 'self' data: https:; connect-src 'self' https:; frame-src 'self' https:;
```

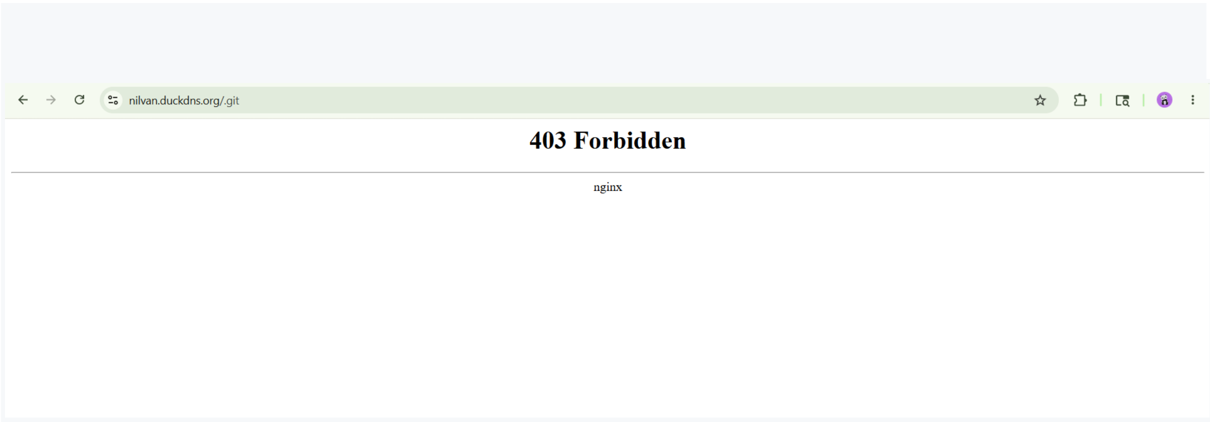
Verify .git Access Blocked

```
curl -I https://nilvan.duckdns.org/.git
```

```
ubuntu@ip-10-0-10-55:~$ curl -I https://nilvan.duckdns.org/.git
HTTP/2 403
server: nginx
date: Sun, 10 May 2026 04:31:07 GMT
content-type: text/html
content-length: 146
x-frame-options: DENY
x-content-type-options: nosniff
referrer-policy: strict-origin-when-cross-origin
strict-transport-security: max-age=63072000; includeSubDomains; preload
permissions-policy: geolocation=(), microphone=(), camera=()
content-security-policy: default-src 'self'; base-uri 'self'; object-src 'none'; frame-ancestors 'none'; form-action 'se
lf'; script-src 'self' 'unsafe-inline' 'unsafe-eval' https:; style-src 'self' 'unsafe-inline' https:; img-src 'self' dat
a: blob: https:; font-src 'self' data: https:; connect-src 'self' https:; frame-src 'self' https:;
```

Expected Output:

```
403 Forbidden
```



17. Conclusion

The Next.js application was successfully deployed on Ubuntu 24.04 with:

- Secure SSH hardening
- Non-root deployment user
- Elastic IP configuration
- GitHub SSH deployment
- PM2 process management
- Nginx reverse proxy
- SSL/TLS using Let's Encrypt
- DuckDNS domain configuration
- Production-ready hardened environment

The deployment follows modern security best practices suitable for production environments